



What is Darwinism?

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In *One Long Argument*, Ernst Mayr (evolutionary biologist, and originator of the Biological Species Concept) summarizes Darwin's theories, and traces the history of their acceptance by the world scientific community.

In the Preface, he begins:

A modern evolutionist turns to Darwin's work again and again. This is not surprising, since the roots of all our evolutionary thinking go back to Darwin. Our current controversies very often have as their starting point some vagueness in Darwin's writings or a question Darwin was unable to answer owing to the insufficient biological knowledge available in his time. But one returns to Darwin's original writings for more than historical reasons. Darwin frequently understood things far more clearly than both his supporters and his opponents, including those of the present day.

In Chapter Four, "Ideological Opposition to Darwin's Five Theories", Mayr summarizes "Darwin's Theory", or "Darwinism", thus:

In both scholarly and popular literature one frequently finds references to "Darwin's theory of evolution", as though it were a unitary entity. In reality, Darwin's "theory" of evolution was a whole bundle of theories, and it is impossible to discuss Darwin's evolutionary thought constructively if one does not distinguish its various components.

... The term "Darwinism", ... has numerous meanings depending on who has used the term and at what period. A better understanding of the meaning of this term is only one reason to call attention to the composite nature of Darwin's evolutionary thought.

... One particular cogent reason why Darwinism cannot be a single monolithic theory is that organic evolution consists of two essentially independent processes, as we have seen: transformation in time, and diversification in ecological and geographical space. The two processes require a minimum of two entirely independent and very different theories.

... I consider it necessary to dissect Darwin's conceptual framework of evolution into a number of major theories that formed the basis of his evolutionary thinking. For the sake of convenience, I have partitioned Darwin's evolutionary paradigm into five theories, but of course others might prefer a different division. The selected theories are by no means all of Darwin's evolutionary theories; others were, for instance, sexual selection, pangenesis, effect of use and disuse, and character divergence. However when later authors referred to Darwin's theory they invariably had a combination of some of the following five theories in mind:

1. **Evolution as such.** This is the theory that the world is not constant or recently created nor perpetually cycling, but rather is steadily changing, and that organisms are transformed in time.
2. **Common descent.** This is the theory that every group of organisms descended from a common ancestor, and that all groups of organisms, including animals, plants, and microorganisms, ultimately go back to a single origin of life on earth.
3. **Multiplication of species.** This theory explains the origin of the enormous organic diversity. It postulates that species multiply, either by splitting into daughter species or by "budding", that is, by the establishment of geographically isolated founder populations that evolve into new species.
4. **Gradualism.** According to this theory, evolutionary change takes place through the gradual change of populations and not by the sudden (saltational) production of new individuals that represent a new type.
5. **Natural selection.** According to this theory, evolutionary change comes about through the

abundant production of genetic variation in every generation. The relatively few individuals who survive, owing to a particularly well-adapted combination of inheritable characters, give rise to the next generation.

Let's look at some of the implications of Mayr's analysis.

At first blush, (4) **Gradualism** seems like it might conflict with Gould & Eldredge's "punctuated equilibrium" theory; but on closer examination, not so.

Here [thanks to Robert Low] are two relevant quotes from *On the Origin of Species*:

"... it is probable that the periods, during which each [species] underwent modification, though many and long as measured by years, have been short in comparison with the periods during which each remained in an unchanged condition." (from the final 6th edition, 1872)

"Varieties are often at first local...rendering the discovery of intermediate links less likely. Local varieties will not spread into other and distant regions until they are considerably modified and improved; and when they do spread, if discovered in a geological formation, they will appear as if suddenly created there, and will simply be classed as new species."

Darwin did not claim that evolutionary change is slow and continuous -- only that it does *not* proceed by "jumps" in a single generation (what Mayr calls "saltational" change). That is, despite the distortions of some anti-evolutionists, Darwin explicitly did not think that evolution proceeds by the production of "hopeful monsters" -- Darwin himself never proposed that a fully-dinosaur parent gave birth to fully-bird progeny. Rather, the change took place in a series of intermediate, perhaps nearly insensible, steps in successive generations. Note that change over a thousand generations of any species appears as "sudden" or "abrupt" change in the fossil record, because a thousand generations is such an infinitesimally small fraction of Earth's history.

(5) **Natural selection**, doesn't account for some of the kinds of variation that we see in species -- particularly non-adaptive traits -- but you'll notice that Darwin didn't claim that natural selection explained all traits, merely the adaptive ones.

After Darwin, some biologists distorted the theory of natural selection into the doctrine of "strict adaptationism", in which every feature of every organism was held to be produced by natural selection (and thus some explanation of why the feature is adaptive was required.) But Darwin didn't say that *all* selection is natural (adaptive) selection -- only that natural selection is the source of *some* change, and can explain why adaptive change occurs. Modern biologists have proposed other mechanisms for change -- neutral selection, genetic drift, the "founder effect", etc. -- and Darwin himself thought that sexual selection could be important. None of these contradict the idea of natural selection; as additional mechanisms for genetic change over time, they augment it.

Here [thanks to Ken Smith] is a quote from the final chapter of the sixth edition of *On the Origin of Species*:

But as my conclusions have lately been much misrepresented, and it has been stated that I attribute the modification of species exclusively to natural selection, I may be permitted to remark that in the first edition of this work, and subsequently, I placed in a most conspicuous position -- namely, at the close of the Introduction -- the following words: "I am convinced that natural selection has been the main but not the exclusive means of modification."

This has been of no avail.

Great is the power of steady misrepresentation; but the history of science shows that fortunately this power does not long endure.

Mayr recaps the history of Darwinist theories, and addresses the claims that Darwinism has been disproved or superseded in Chapter Ten: "New Frontiers in Evolutionary Biology".

Just as in the decade after the rediscovery of Mendel's rules, since about 1970 the claim has been made increasingly often that "Darwinism is dead."

...

Opponents of the [modern evolutionary] synthesis consistently confound three schools of Darwinism:

1. neo-Darwinism, a term coined by Romanes in 1896 to designate "Darwinism without an inheritance of acquired characters";
2. early population genetics, a strongly reductionist school that defined evolution as the modification of gene frequencies by natural selection; and

3. the holistic branch of the [modern evolutionary] synthesis, which continued the traditions of Darwin and the naturalists while accepting the findings of genetics.

...

Darwinism is not a simple theory that is either true or false but is rather a highly complex research program that is being continuously modified and improved. This was true before the [modern evolutionary] synthesis, and it continues to be true after the synthesis. Table 2 lists many of the significant stages in the modification of Darwinism that one might recognize. Yet recognizing such seemingly discontinuous periods is in many respects an artificial enterprise. ... each of these periods was heterogeneous to some extent, owing to the diversity in the thinking of different evolutionists. Most critics who have attempted to refute the evolutionary synthesis have failed to recognize this diversity of views and thus have succeeded in refuting only the reductionist fringe of the Darwinism camp.

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Table 2
Significant stages in the modification of Darwinism

<u>Date</u>	<u>Stage</u>	<u>Modification</u>
1883-1886	Weismann's neo-Darwinism	End of soft inheritance; diploidy and genetic recombination recognized
1900	Mendelism	Genetic constancy accepted and blending inheritance rejected
1918-1933	Fisherism	Evolution considered to be a matter of gene frequencies and the force of even small selection pressures
1936-1947	Evolutionary synthesis	Population thinking emphasized; interest in the evolution of diversity, geographic speciation, variable evolutionary rates
1947-1972	Post-Synthesis	Individual increasingly seen as target of selection; a more holistic approach; increased recognition of chance and constraints
1954-1972	Punctuated equilibria	Importance of speciation evolution
1969-1980	Rediscovery of importance of sexual selection	Importance of reproductive success for selection

References

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On the Origin of Species by Means of Natural Selection, or the Preservation of Favored Races in the Struggle for Life, Charles Darwin, First Edition 1859. Sixth Edition 1872.

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